

## Case Study: Ransomware

### What Is Ransomware?

Ransomware is a type of malicious software (malware) that encrypts a victim's files or locks them out of their system, after which attackers demand a ransom payment — typically in cryptocurrency — in exchange for restoring access.

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### How It Typically Happens (Step-by-Step)

1. **Initial Infection** – The attacker sends a phishing email containing a malicious link or attachment. The victim clicks it, unknowingly downloading the ransomware.
  2. **Execution** – The malware silently installs itself and begins scanning the system for valuable files (documents, databases, photos).
  3. **Encryption** – All targeted files are encrypted using strong cryptographic algorithms, making them completely inaccessible to the user.
  4. **Ransom Demand** – A message appears on screen demanding payment (often Bitcoin) within a deadline, threatening permanent deletion of files if unpaid.
  5. **Payment/No Recovery** – Even if the ransom is paid, attackers may not always restore access.
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### Who Is Usually Targeted?

- Hospitals and healthcare institutions
  - Government agencies
  - Schools and universities
  - Small-to-medium businesses with weak cybersecurity
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### Consequences

- **Financial loss** – Ransom payments can range from thousands to millions of dollars.
  - **Data breach** – Sensitive personal or organisational data may be leaked publicly.
  - **Operational disruption** – Services shut down for days or weeks.
  - **Reputational damage** – Loss of public trust, especially for institutions handling sensitive data.
  - **Legal liability** – Organisations may face penalties for failing to protect user data.
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